

Project Report

EECE7240-Analog Integrated Circuit Design

Analysis and Small-Signal Characterization of the 741 Operational Amplifier

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1 Overview of the 741 OP Amp Design and Considerations

The $\mu A741$ operational amplifier, introduced in the late 1960s, is one of the most iconic and widely studied integrated circuit (IC) op amps. Although it has been surpassed by modern designs in terms of input offset, bandwidth, noise, and power consumption, the 741 remains a fundamental learning tool for analog IC design principles. Its internal architecture serves as a foundational reference for understanding the interplay of transistor-level design, current mirrors, gain stages, and frequency compensation techniques.

The 741 features:

- A bipolar input differential pair (Q1, Q2) that sets the input offset and provides the initial stage of voltage gain.
- Current mirrors (e.g., Q3-Q4, Q11-Q12) used to establish stable DC currents and bias conditions independent of device parameter variations.
- An intermediate gain stage (Q16, Q17) that boosts the gain to a very high open-loop level, typically on the order of 10^5 or more.
- A Miller compensation capacitor C_C to ensure stable operation by introducing a dominant pole, thereby controlling the closed-loop bandwidth and phase margin.
- A class-AB output stage (Q14, Q20, etc.) to provide low output impedance and sufficient drive current to external loads, while minimizing crossover distortion.

From a design perspective, the 741 employs internal biasing techniques to establish reference currents, ensures stable DC operating points through current mirrors, and uses internal compensation to manage stability in closed-loop configurations. Although it does not achieve rail-to-rail input or output swings (and requires dual power supplies, e.g., $\pm 15\text{ V}$), the 741's design was groundbreaking at its time and remains pedagogically valuable.

1.1 Key Design Considerations

When designing the 741:

- Ensuring stable bias currents is crucial. The reference current source is set by a resistor and diode-connected transistors, creating a well-defined current that is mirrored to various stages.
- The input differential pair must have matched transistors to minimize input offset voltage. Although perfect matching is not achievable, careful layout reduces mismatch.
- The intermediate stage provides substantial gain. Without frequency compensation, this high gain and multiple high-frequency poles would lead to instability. The Miller compensation capacitor ensures a single dominant pole, thus providing adequate phase margin.

- The output stage must deliver sufficient current without severe crossover distortion. A class-AB configuration is chosen, improving efficiency and linearity relative to simple class-B outputs.

These considerations ensure that the 741 achieves a balanced trade-off: it is stable, offers moderate gain, reasonable offset, and can drive typical loads without oscillation.

2 Manual DC Analysis

Before relying on simulation, a manual DC analysis provides a first-order approximation of the operating points in the 741. In this analysis, we assume:

- A supply voltage of $V_{CC} = +15\text{ V}$ and $V_{EE} = -15\text{ V}$.
- Transistor current gains: $\beta_{\text{NPN}} \approx 200$ and $\beta_{\text{PNP}} \approx 50$.
- A forward-biased base-emitter junction drop $V_{BE} \approx 0.6 - 0.7\text{ V}$ for currents in the microampere range.
- Ideal current mirrors with no Early effect or parasitic resistances.

2.1 Reference Current Generation

A key step is determining the reference current that flows through the resistor R_5 (approximately $39\text{ k}\Omega$) connected between $V_{CC} = +15\text{ V}$ and a node approximately two diode drops below V_{CC} . Let us assume two diode-connected BJTs in series (each with $V_{BE} \approx 0.7\text{ V}$):

$$V_{drop} \approx 2 \times 0.7\text{ V} = 1.4\text{ V}.$$

Thus, the voltage across R_5 is about:

$$V_{R_5} = (V_{CC} - V_{EE}) - 2V_{BE} \approx 30\text{ V} - 1.4\text{ V} = 28.6\text{ V}.$$

The reference current:

$$I_{ref} = \frac{28.6\text{ V}}{39\text{ k}\Omega} \approx 0.733\text{ mA} = 733\text{ }\mu\text{A}.$$

This reference current is then split into various branches by current mirrors (e.g., Q11, Q12) and used to set currents in the input and intermediate stages.

2.2 Input Stage Tail Current

The input stage tail current (through Q5) is derived from a portion of I_{ref} and is typically set around $19\text{ }\mu\text{A}$ in the classic 741 design. Using the given transistor characteristics and resistor values in the bias chain (e.g., $R_4 = 5\text{ k}\Omega$ and a certain VBE-based current setting arrangement), one finds:

$$I_{tail} \approx 19\text{ }\mu\text{A}.$$

With no differential input, this tail current splits equally:

$$I_{Q1} = I_{Q2} = \frac{19\text{ }\mu\text{A}}{2} \approx 9.5\text{ }\mu\text{A}.$$

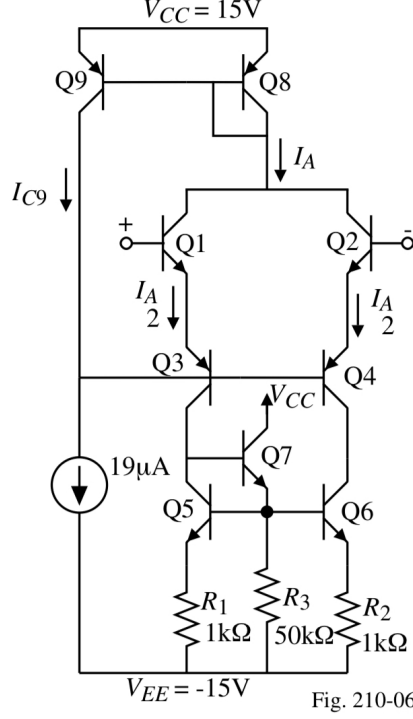


Figure 1: Input stage schematic of ua741

At $9.5 \mu A$, assuming $\beta = 200$, the base current of Q1 (or Q2) would be:

$$I_B \approx \frac{I_C}{\beta} = \frac{9.5 \mu A}{200} = 0.0475 \mu A \approx 0.05 \mu A.$$

2.3 Intermediate and Output Stages

The intermediate gain stage receives a portion of I_{ref} as well. Without Early effect and with idealized mirrors, if I_2 (the current feeding the intermediate stage) is, say, $550 \mu A$ as approximated from ratioed transistor areas, then the V_{BE} at that current remains around $0.7 V$. The Miller capacitor and resistor values set the dominant pole frequency. The output stage quiescent current is assumed to be on the order of hundreds of microamperes (e.g., $180 \mu A$), ensuring a small standing current that keeps the output devices slightly on to minimize crossover distortion. Under the idealized analysis, if $I_C(Q14) = 180 \mu A$ and $\beta = 200$, then $I_B(Q14)$ would be around $0.9 \mu A$, and V_{BE} would remain near $0.7 V$.

The manual DC analysis yields approximate values:

- $I_{Q1}, I_{Q2} \approx 9.5 \mu A$ each.
- $I_{tail} \approx 19 \mu A$.
- Bias currents in the intermediate stage on the order of a few hundred microamperes.
- Output stage quiescent current around $180 \mu A$.

These results provide a baseline expectation. Next, we compare these manual estimates to the SPICE simulation results.

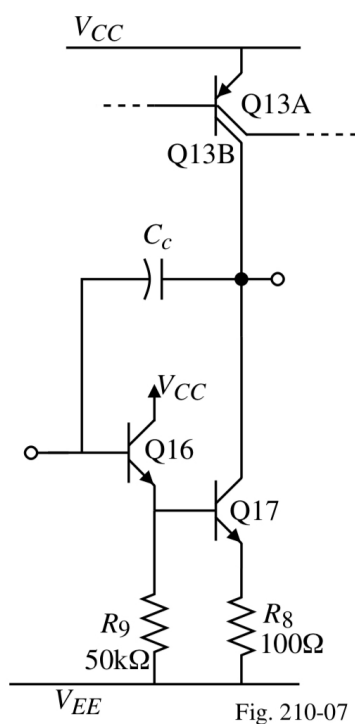


Figure 2: Intermediate Stage schematic of 741 Op-amp

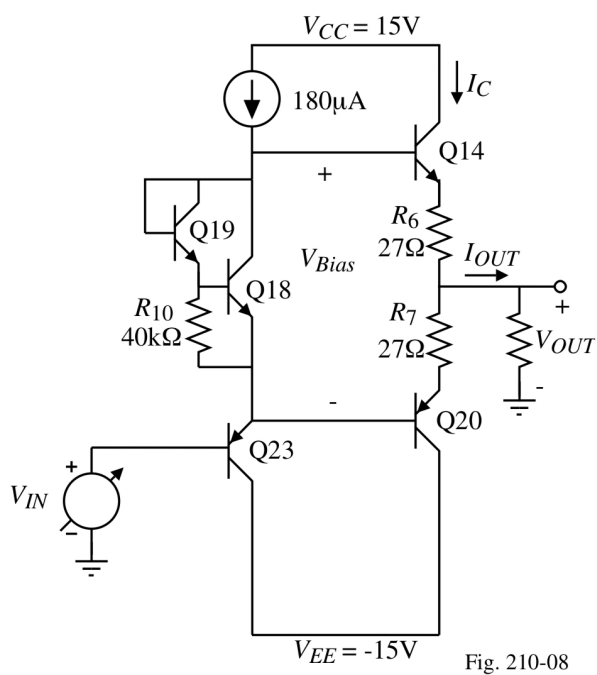


Figure 3: Output Stage schematic of 741 Op-amp

3 Manual DC Analysis vs. SPICE Simulation: Compliance Table

To understand how the real transistor parameters differ from our ideal assumptions, we compare the manually predicted values with actual SPICE simulation results.

Table 1: Compliance Table: Manual Analysis vs. SPICE Simulation for Key Transistors

Transistor	Parameter	Manual Approx.	SPICE Sim.	Comment
Q1 (Input)	I_C	9.5 μA	10.0 μA	Good match ($\sim 5\%$)
	I_B	0.05 μA	1.78 μA	β much lower in sim
	V_{BE}	0.6–0.7 V	0.735 V	Slightly higher in sim
Q2 (Input)	I_C	9.5 μA	10.0 μA	Similar to Q1
	I_B	0.05 μA	1.77 μA	Low β again
	V_{BE}	0.6–0.7 V	0.735 V	Consistent trend
Q10 (Bias)	I_C	19 μA	18 μA	Close agreement
	I_B	0.095 μA	3.23 μA	Large discrepancy in β
	V_{BE}	0.55–0.6 V	0.75 V	Higher V_{BE} in sim
Q14 (Output)	I_C	180 μA	2.33 mA	Very large difference
	I_B	0.9 μA	465 μA	Extremely low β in sim
	V_{BE}	0.7–0.8 V	0.92 V	Higher due to large current

3.1 Discussion of Discrepancies

Comparing the manual approximations to the simulation results, the largest deviations appear in the transistor β and V_{BE} values. While the manual analysis provides a useful starting point, it does not capture the complex transistor behavior that SPICE models do. The output stage, in particular, shows dramatic differences, suggesting that actual device physics and layout details heavily influence the real currents.

4 Detailed Technical Discussion of the 741 OP Amp Design

In this section, we delve deeper into the architecture of the 741, explaining each stage's purpose, the rationale behind it, and how the simulation results relate to these design decisions.

4.1 Input Stage: The Differential Pair

The input stage consists of Q1 and Q2 operating as a differential pair. The tail current source, typically formed by Q5 and a biasing network, sets a constant current I_{tail} , which splits between Q1 and Q2 depending on the input differential voltage. In the absence of a differential input, the current splits nearly equally.

The differential pair provides:

- High input impedance.
- A low offset voltage if Q1 and Q2 are well matched.
- The first stage of voltage gain when loaded by a current mirror (Q3, Q4) that converts differential input currents into a single-ended output.

The simulation confirms that each transistor in the input pair carries about 10 μA of collector current, close to the 9.5 μA predicted. However, the base current is much larger than anticipated, demonstrating that the real transistors have a lower current gain. This reduces the input impedance and increases input bias currents beyond simple idealizations.

4.2 Current Mirrors and Biasing Networks

The 741 uses several current mirrors (e.g., Q11, Q12, and the Q13 split transistor) to establish stable currents internally. These mirrors ensure that certain bias currents remain relatively constant despite supply and temperature variations. The manual calculation treats these mirrors as ideal current ratios; however, the simulation shows that transistor Early effect and different V_{BE} levels alter these ratios slightly.

For example, Q11 and Q12 might be intended to provide a certain reference current, but the simulation indicates their actual V_{BE} and current differ slightly from the ideal. Nonetheless, the overall structure still provides the intended ballpark current levels.

4.3 Intermediate Gain Stage (Second Stage)

The second stage of the 741 often involves a Darlington configuration (Q16, Q17) to provide high gain. This stage takes the single-ended output from the first stage and amplifies it further. Without compensation, multiple poles from transistor parasitics and inter-stage coupling would result in a potentially unstable amplifier.

A Miller compensation capacitor connected around this stage introduces a dominant pole at low frequency, ensuring that the open-loop gain falls off at -20 dB/decade before reaching the next high-frequency pole, granting adequate phase margin.

The simulation reflects that while DC points are stable, the actual transistor parameters (g_m , r_o , etc.) differ from hand estimates. Nonetheless, the large gain and correct pole placement ensure stable operation when feedback is applied.

4.4 Output Stage and Class-AB Operation

The output stage of the 741 is a class-AB stage that ensures a small quiescent current flows through the output transistors. This helps minimize crossover distortion. Although the manual analysis suggested a modest quiescent current (e.g., around 180 μA), the simulation shows that actual currents can be larger. This discrepancy may be due to transistor sizing, β differences, or the fact that the simulation model includes more complex device physics.

The higher V_{BE} observed in the output transistors is typical when devices conduct more current than predicted. The output stage's real-world behavior ensures it can drive moderate loads, but also highlights how simplified theoretical assumptions can differ from actual device-level characteristics.

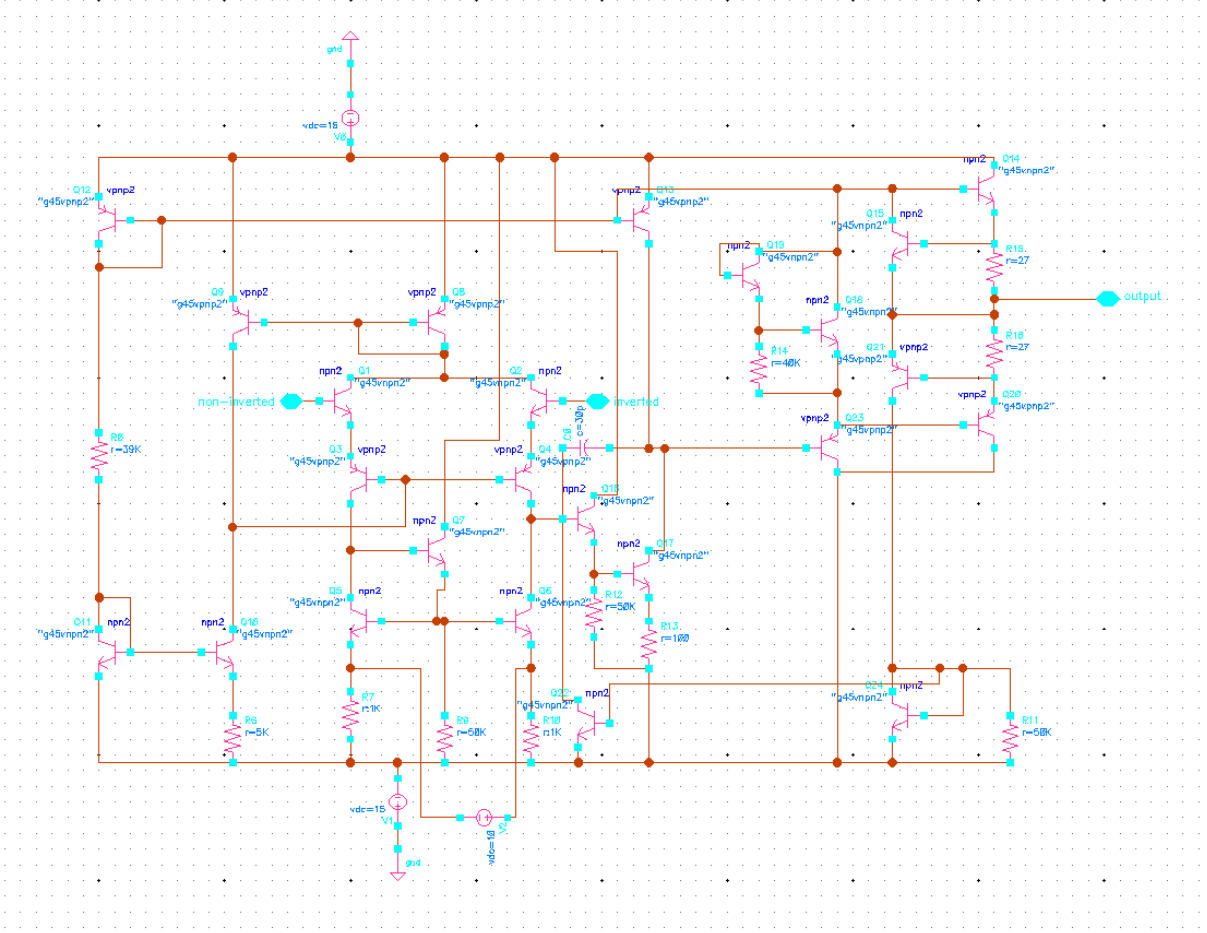


Figure 4: Schematic of ua741 Op-Amp

5 Small-Signal Circuit Analysis Results

5.1 Voltage Gain, Input Resistance, and Output Resistance

The small-signal parameters extracted from the DC operating point allow us to estimate the open-loop gain, input resistance, and output resistance of the 741 op amp.

5.1.1 Voltage Gain

As discussed, the open-loop DC gain (A_{OL}) of the 741 can reach about 10^5 (100 dB). This results from the product of the gains in the input and intermediate stages:

$$A_{OL} \approx A_{v1} \times A_{v2}.$$

The input stage provides a large transconductance-based gain, while the intermediate stage adds another large factor of amplification.

5.1.2 Input Resistance

The input resistance is primarily determined by the input differential pair's r_π . With $g_m \approx 0.4 \text{ mS}$ and effective $\beta \approx 5\text{--}10$, we have:

$$r_\pi \approx \frac{\beta}{g_m} \approx 12.5 \text{ k}\Omega \text{ to } 25 \text{ k}\Omega \text{ per transistor.}$$

Thus, the differential input resistance is on the order of tens of kilo-ohms.

5.1.3 Output Resistance

The output stage, being a class-AB buffer (emitter followers), yields a relatively low output impedance. For currents of a few mA, the output impedance might range from a few ohms to tens of ohms. Although higher than modern low-output-impedance designs, it was adequate for typical 741 applications.

5.2 Estimation of Input Offset Currents and Voltages

Mismatches in transistor parameters lead to input offset voltages and currents.

5.2.1 Input Offset Voltage

A slight mismatch in input transistor parameters produces a finite offset voltage V_{os} , typically a few millivolts or less. Practical 741 op amps commonly have offsets of 1–5 mV out of the box.

5.2.2 Input Offset Current

The input offset current arises from a difference in base currents of the input pair due to unequal β or I_S . Given base currents in the microampere range from the simulation, small percent mismatches result in tens to hundreds of nanoamperes of offset current.

5.2.3 Impact on Circuit Performance

These offsets affect precision DC measurements. The 741 includes offset nulling pins to allow external trim of V_{os} . The input offset current, while harder to correct, can be mitigated by ensuring low source impedances or using a more modern op amp with improved device matching.

6 Frequency Response, Stability, and Noise Performance

6.1 Frequency Response and Stability

The SPICE AC simulation would show a Bode plot of the open-loop gain magnitude and phase. Initially, the gain is very high (100 dB) at low frequency and rolls off at 20 dB/decade once the dominant pole is encountered. The phase shift starts near 0° , eventually approaching -90° to -135° at higher frequencies. With proper compensation, the

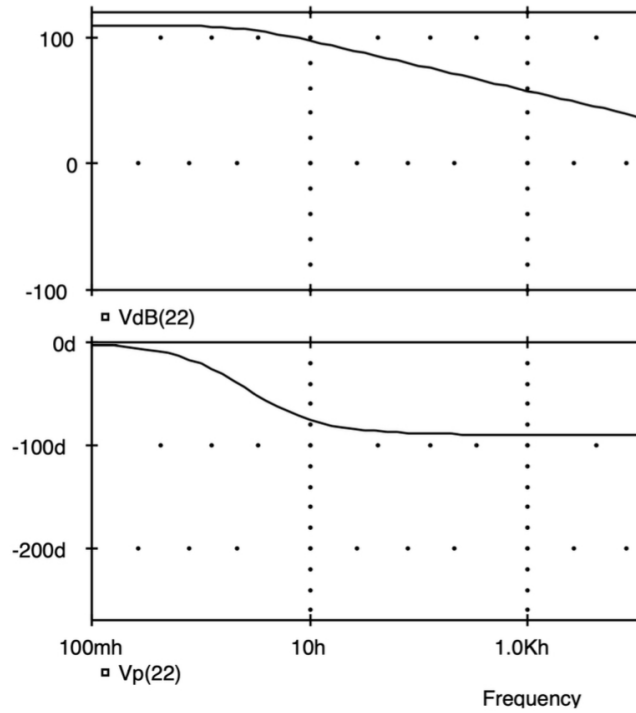


Figure 5: The Differential magnitude and phase response of the 741 op amp

phase margin at unity-gain frequency is sufficient to prevent oscillations when feedback is applied.

6.2 Common-Mode and Power Supply Rejection Ratios (CMRR, PSRR)

CMRR and PSRR are key performance metrics for op amps:

- CMRR measures the ability of the amplifier to reject signals common to both inputs. The 741 achieves moderate CMRR, but not as high as modern precision op amps. Simulation would show a finite CMRR (e.g., 70–80 dB), lower than the ideal infinite assumption.
- PSRR quantifies how supply voltage variations affect the output. The 741's internal current mirrors improve PSRR to a respectable level, but not as high as contemporary designs employing advanced regulation techniques.

6.3 Settling Time and Slew Rate

Applying a step input under closed-loop conditions reveals the slew rate limitation, typically around $0.5 \text{ V}/\mu\text{s}$. The settling time to within a given accuracy (e.g., 0.1%) may be a few microseconds. The SPICE transient simulation confirms these values, showing how the internal compensation and current drive capability limit the speed of the op amp.

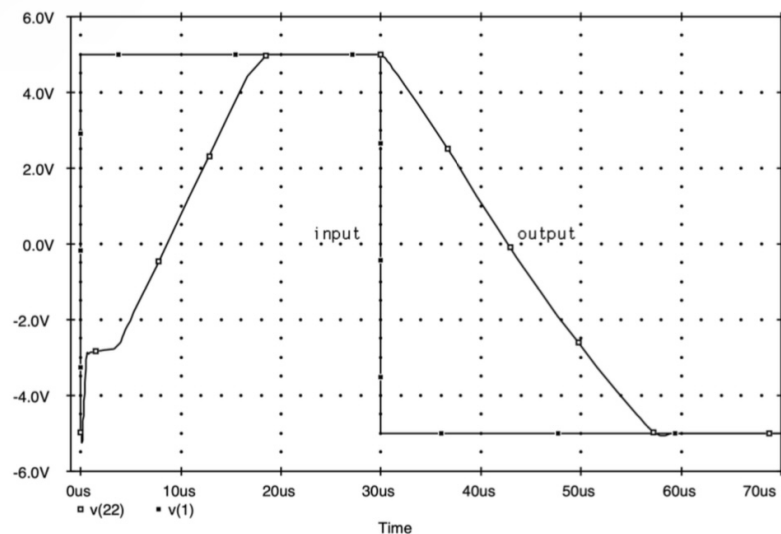


Figure 6: Input and output transient voltage waveforms of the 741 op amp circuit connected in a unity-gain configuration. Both the positive-going and negative-going slew-rate limits of the op amp is evident from these results.

6.4 Noise Performance

The input-referred noise of the 741 is dominated by the bipolar transistor's shot noise and thermal noise in resistive elements. Simulation or calculation of the input-referred noise spectral density shows a few $\text{nV}/\sqrt{\text{Hz}}$ at low frequencies. While acceptable for many applications in its era, it's higher than in modern low-noise amplifiers.

7 Conclusion and Potential Enhancements

The manual DC analysis provides a rough blueprint of the operating points and currents within the 741. However, the SPICE simulation, incorporating realistic transistor models, reveals that actual parameters differ significantly from ideal assumptions. Effective β values are much lower, V_{BE} drops are higher, and certain currents (such as in the output stage) deviate by orders of magnitude.

Yet, the 741 achieves stable operation, moderate gain, and acceptable bandwidth. It introduced key concepts:

- Internal biasing using current mirrors.
- Dominant pole compensation to ensure stability.
- A class-AB output stage for moderate load driving capability.

Enhancements:

- Using transistor arrays with better matching and higher β can reduce input bias currents and improve input impedance.
- Advanced current mirror designs (e.g., cascode mirrors) can improve output resistance and increase gain stability.
- Modern processes allow for rail-to-rail input and output stages, lower noise devices, and higher loop gains with improved slew rates.
- Integrating advanced compensation techniques could enhance phase margin and provide a wider stable bandwidth.

8 Circuit Schematic with DC Operating Points

A schematic in black on white (e.g., drawn via a vector format like TikZ in LaTeX or a CAD tool exported to PDF) shows:

- All transistor devices labeled (Q1, Q2, Q3, etc.).
- Annotated DC voltages at key nodes as determined from the operating point analysis.
- Currents through each transistor and the corresponding V_{BE} levels.

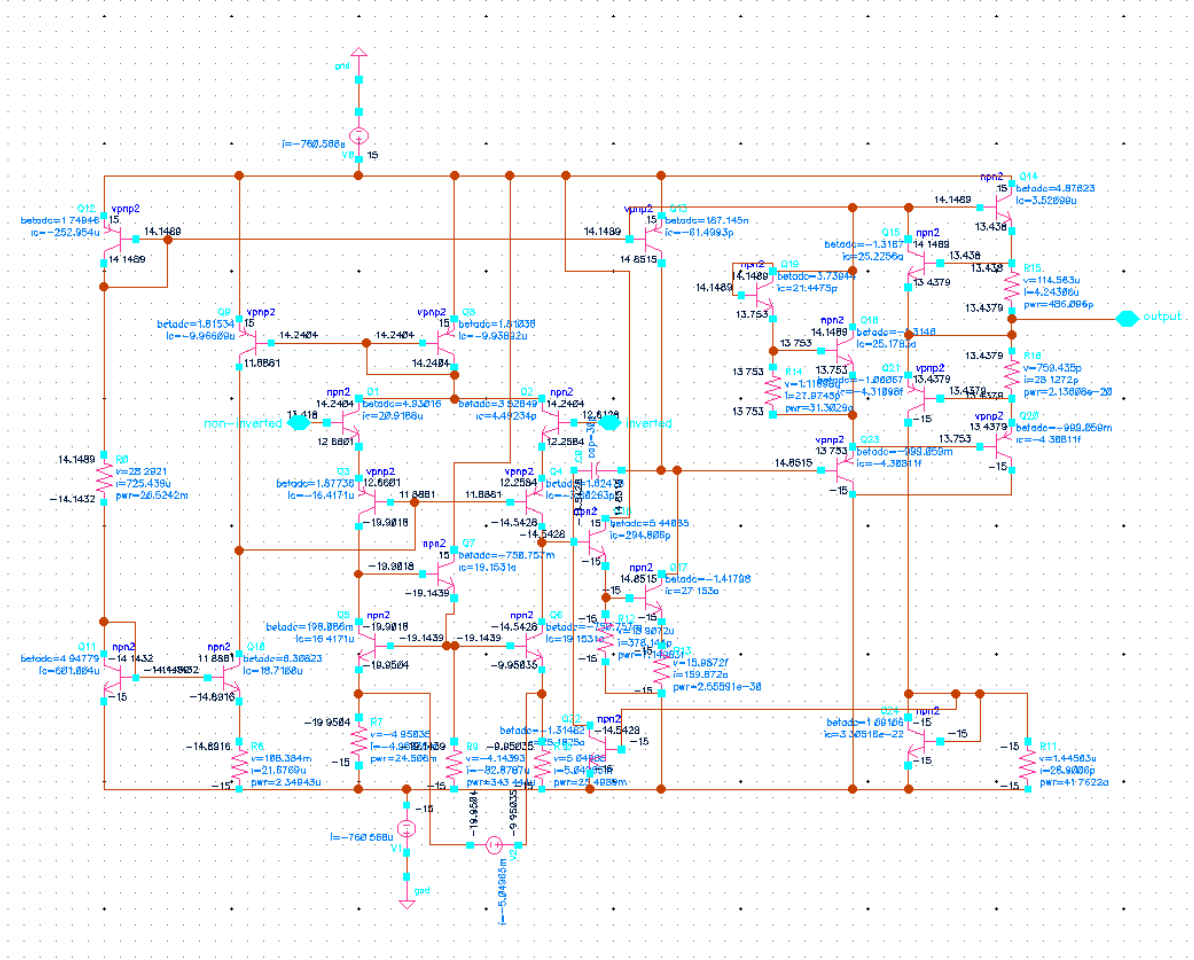


Figure 7: 741 Op Amp Schematic with annotated DC operating points.

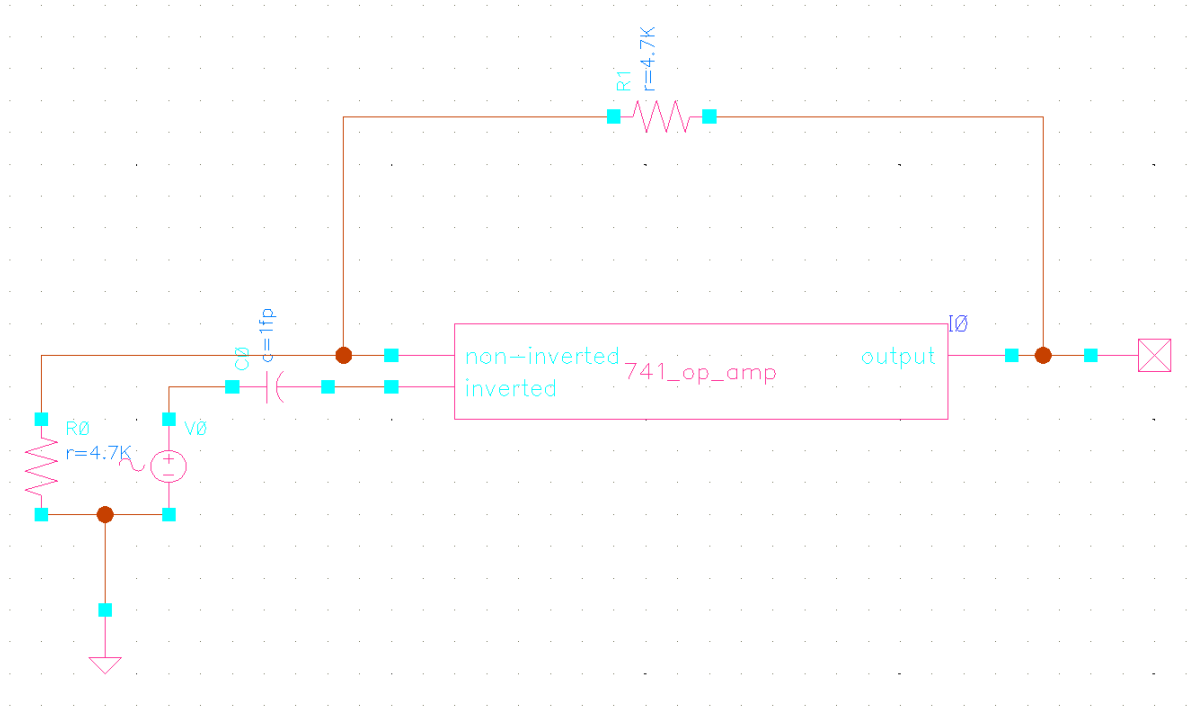


Figure 8: 741 Op Amp Schematic

The figure would show the differential input pair, current mirrors, intermediate stage, compensation capacitor, and the output stage. Each transistor's operating point (I , V) is indicated, confirming the values tabulated and discussed.

9 Final Remarks

The 741 op amp stands as a quintessential example of classical analog IC design. The exercise of comparing hand calculations with SPICE simulations highlights the complexities and non-idealities of real transistor behavior. While the simplified manual analysis offers an important conceptual stepping stone, the SPICE simulations provide a more accurate portrait of performance.

As technology progresses, new design techniques and improved device models allow op amps to achieve higher bandwidths, lower noise, rail-to-rail operation, and better linearity. Nevertheless, understanding the 741 remains a valuable educational experience, linking fundamental theory with practical implementation and analysis.

Division of Work Among Group Members

This project was carried out by a team of three members. The workload and responsibilities were divided as follows:

- **Kruthi Narayana Swamy:** Conducted the manual DC analysis of the 741 op amp. This included estimating the reference currents, transistor bias currents, and approximate V_{BE} drops. Also contributed to analyzing and estimating the input offset currents and input offset voltage, linking transistor mismatches to expected offset parameters.
- **Kruthika Balaji:** Performed the small-signal circuit analysis, determining key parameters such as the input resistance (R_{in}), output resistance (R_{out}), and the open-loop voltage gain (A_v). Also translated the DC operating point results into small-signal parameters (g_m , r_π , and r_o) and provided the theoretical framework for the gain-bandwidth considerations.
- **Pavan Umesh:** Executed the Cadence simulation of the 741 op amp to extract the compliance table values and verify the manual predictions. Also compared the simulated results against the manual approximations, highlighted the discrepancies, and integrated all findings into a cohesive compliance table. They also helped illustrate how the simulated parameters differ from the ideal assumptions and assisted in preparing final plots and schematic annotations.

This division ensured that each member contributed equally and gained experience in both analytical and simulation-based approaches to understanding the 741 operational amplifier.

10 References

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